



Clostridium Difficile Infections and Treatment

Overview of Clostridium Difficile Infections (CDI)

Clostridioides difficile infection (CDI) remains a significant clinical challenge both in the management of severe and severe-complicated disease and the prevention of recurrence.

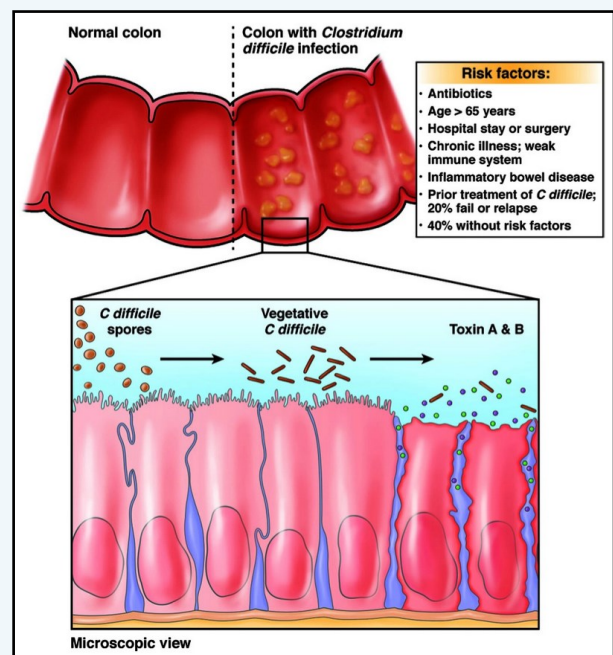
Most of the cases we will encounter will be the non-severe type and are amenable to treatment over the phone. The diagnosis of CDI is usually considered when the patient has had 3 or more watery stools in 24 hours.

Risk Factors for CDI

- Antibiotic exposure (especially fluoroquinolones, third or fourth generation cephalosporins, clindamycin, carbapenems)
- Gastrointestinal surgery or manipulation
- Long length of stay in healthcare settings
- A serious underlying illness
- Immunocompromising conditions
- Advanced age

Clinical Features of CDI

- Watery diarrhea
- Fever
- Loss of appetite
- Nausea
- Abdominal pain or tenderness



Note. From *C. diff* [Infographic], by AGA GI Patient Center, n.d., AGA GI Patient Center (<https://patient.gastro.org/clostridioides-difficile-c-diff/>)

Classification of Severity of CDI

Severity	IDSA/SHEA 2021	ESCMID 2021	ASID 2016
Non-severe	WBC count of ≤ 15000 cells/mL and a serum creatinine level < 1.5 mg/dL.	WBC count of ≤ 15000 cells/mL and a serum creatinine level $\leq 50\%$ above baseline, and core body temperature at presentation $\leq 38.5^\circ\text{C}$. No imaging features of severity.	Absence of all features consistent with severe CDI.
Severe	One of the following factors at presentation: WBC count of > 15000 cells/mL or a serum creatinine level ≥ 1.5 mg/dL.	One of the following factors at presentation: WBC count of > 15000 cells/mL or a rise in serum creatinine level $> 50\%$ above baseline or core body temperature $> 38.5^\circ\text{C}$. Additional supporting factors, when available, are distension of the large intestine, pericolic fat stranding or colonic wall thickening (including low-attenuation mural	Any of the following features if no other explanation can be provided: WBC count of > 15000 cells/mL or a rise in serum creatinine level $> 50\%$ above baseline or core body temperature $> 38.5^\circ\text{C}$. Rigors, haemodynamic instability, peritonitis or evidence of bowel perforation, ileus or toxic megacolon, elevated lactate level, albumin level < 25 mg/L, large intestine distension, colonic wall thickening, fat stranding, unexplained ascites (imaging) or pseudomembranous colitis on
Severe-complicated 'fulminant'	Presence of hypotension or shock, ileus or megacolon.	Presence of one of the following factors that needs to be attributed to CDI: Hypotension, septic shock, elevated serum lactate, ileus, toxic megacolon, bowel perforation or any fulminant course of disease (i.e., rapid deterioration of the patient).	An episode of CDI complicated by: Toxic megacolon, admission to intensive care for severe sepsis, requirement for surgery, or death due to CDI.

We should only be treating "non-severe" CDI infection during a Get Care Now visit. If they have any of the above severe symptoms, then the patient should be evaluated in person.

CDI Treatment

The KP Gold Card outlines the recommended treatment for CDI.

Initial Episode

Vancomycin 125 mg PO QID x 14 days

First Recurrence

Vancomycin taper is as follows (with capsules):

- 125 mg PO QID x 14 days,
- Then 125 mg PO BID x 7 days,
- Then 125 mg PO daily x 7 days,
- Then 125 mg PO every other day x 7 days,
- Then 125 mg PO every 3 days x 14 days



Note. From *Vancomycin Hydrochloride* [Photograph], by eMedic Store, n.d., eMedic Store (<https://emedicstore.org/products/Vancomycin-250mg-Tablet>)

Second (or Later) Recurrence

Review with ID or GI Specialist for other approaches.

If Patient Cannot Take Vancomycin

For an initial episode of mild-to-moderate infection when patient is unable to tolerate or take vancomycin:

Metronidazole * 500 mg PO TID x 10-14 days

*Note: Metronidazole is less effective than other options.

Pediatric Treatment for CDI

*Antimicrobial therapy is **NOT** indicated in children with asymptomatic colonization with *C. difficile*.

Treatment Options for Symptomatic Children

1. Vancomycin (40mg/kg/day) divided QID (max dose 125mg/dose) x 10 days, or
2. Metronidazole (30mg/kg/day) divided QID (max dose 500mg/dose) x 10 days

Other Notes

- Proton pump inhibitors, H2-blockers, and/or use of high-risk/repeated ABX courses may increase risk.
- Advise patients to promptly report recurrent symptoms.
- Do **NOT** prescribe probiotics during an active *C. difficile* infection.
- Consider inpatient treatments for severe, complicated disease.
- If the KP lab receives an order for *C. diff* testing and receives a sold stool, the lab result will say "test not performed" due to the low probability of *C. diff* in a solid stool sample.

References

- AGA GI Patient Center. (n.d.). *C. diff*. [Infographic]. AGA GI Patient Center. <https://patient.gastro.org/clostridioides-difficile-c-diff/>
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